reconnaissance, electronic warfare, and precision strike missions in contested environments.

Featuring a flying-wing configuration reminiscent of the US B-2 Spirit stealth bomber, the CH-7 utilizes radar-absorbent composite materials and a smooth, tailless design to minimize radar cross-section (RCS). It measures approximately 10 meters in length with a 22-meter wingspan, optimized for high-altitude, long-endurance operations, reportedly capable of cruising at over 800 km/h with a ceiling exceeding 40,000 feet (Slocombe, 2019).

The UAV incorporates autonomous flight control, sensor fusion, and AI-driven mission management systems, enabling it to operate independently or in coordinated missions with other drones or manned aircraft. Its internal weapons bay allows for the deployment of precision-guided munitions, anti-radiation missiles, and electronic warfare payloads while maintaining a stealth profile.

Strategically, the CH-7 Rainbow symbolizes China's ambition to compete in the global stealth UAV domain, aligning with its broader goal of network-centric warfare and multi-domain intelligence superiority. Its integration of AI, stealth design, and electronic warfare capability positions it as a vital component in China's next-generation air combat strategy.

Figure 5.4 illustrates the CH-7 Rainbow (2019), a stealth unmanned combat aerial vehicle (UCAV) developed by the China Aerospace Science and Technology Corporation (CASC). The figure highlights the CH-7's flying-wing configuration, designed for low radar cross-section and aerodynamic efficiency, a defining feature of modern stealth aircraft. The smooth, tailless structure and shielded engine intakes minimize both radar and infrared signatures, allowing for deep-penetration missions in contested airspace.